MFE5340 AI in FE: Quantitative Investment
(Spring 2026)

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1. Course Description

This course provides a comprehensive introduction to machine learning in quantitative investment, bridging traditional econometrics and modern machine learning techniques. Core topics include asset pricing theory, high-dimensional regression, PCA and factor models, and multiple hypothesis testing. Advanced machine learning applications, such as tree-based methods, neural networks, and deep learning, are also covered. Students will learn to tackle real-world challenges like data interpretability, missing data, and model complexity. With a focus on practical applications in finance, this course equips students to analyze complex financial data and apply machine learning in quantitative investment strategies.

2. Lectures & Office Hours

Lectures (3 hours per week):
Wednesday 6:00 – 9:00 PM, TD103

Office Hours:
Chuan Shi: Wednesday 4:00 – 5:00 PM (DY520), or by appointment
Zihao Zhu: TBD
Wanting Yang: TBD

3. Prerequisites

Undergraduate level knowledge of probability, statistics, linear algebra, and econometrics.

Familiarity with Python (assignments will require coding with Python).

Basic knowledge of factor models is beneficial.

4. Learning Outcomes

Knowledge Outcomes:

- Apply machine learning in quantitative investment with guidance of financial theory.
- Apply machine learning and econometric methods to analyze financial data and interpret model performance.
- Understand and implement multiple hypothesis testing within financial research contexts.
- Conduct empirical studies and develop replicable, data-driven investment strategies.
- Gain practical knowledge on applying machine learning techniques to real-world financial scenarios.

Skills Outcomes:

- Utilize Python to apply econometric and machine learning techniques in quantitative investment.
- Effectively communicate findings from financial data analysis to both technical and non-technical audiences.
- Design and implement robust backtesting frameworks to assess the performance and risk of machine learning-based strategies.

Values Outcomes:

- Foster enthusiasm and a rigorous analytical mindset for applying machine learning in finance.
- Recognize the value of combining theoretical finance and statistical knowledge with practical coding skills.
- Develop a critical understanding of the limitations and assumptions underlying machine learning models.
- Stay mindful of potential biases like data snooping, cultivating an objective, impartial approach in financial research.

5. Course syllabus

Week	Content/ topic/ activity
1	Introduction to Quantitative Investment Overview of Quantitative Investment Importance of Factor Investing in Modern Asset Management
2	Asset Pricing Theory Fundamental Concepts (SDF, Mean-Variance Analysis, CAPM, APT) Review of Econometric Methods
3	Transition from Econometrics to Machine Learning High-dimensionality of Covariates / Nonlinear Relationships A New Norm in the Era of Big Data
4	Machine Learning in Asset Pricing Properties of Financial Data Overview of Machine Learning Fact and Fictions about Machine Learning in Quantitative Investment
5	Multiple Hypothesis Testing Overview of Multiple Hypothesis Testing Minimum Bayes Factor Double Bootstrap Approach
6	Linear and Penalized Regressions Ridge and Lasso Regressions Elect net
7	PCA and Latent Factor Models (I) PCA and Least Squares Perspective Pisk-Premium PCA
8	Midterm (tentative)
9	PCA and Latent Factor Models (II) General PCA Framework Instrumented PCA, Kernel Trick
10	Advanced Machine Learning Applications Tree-based Methods Neural Networks and Deep Learning
11	Alternative Data and Its Application Overview Extract Stock Return Information from Text Data
12	Interpretability and Missing Data Feature Importance Missing Data
13	Model Complexity and Investor Learning Virtue of Complexity? Machine Learning as Model of Investor Belief Formation
14	Project Presentation Final Project Presentation

6. Assessment Scheme

Component/ method	% weight
Assignments	20
Mid-term Exam	40
Final Exam / Project	40

7. Grade descriptor

Grade	Description
A	Outstanding understanding of the theory and methodology of machine learning in quantitative investment and exceptional application of techniques to real-world financial data using Python. Demonstrates an advanced ability to model and interpret financial patterns and risks using machine learning methods, strongly exceeding expectations for high-quality research in empirical asset pricing and for industry roles.
A-	Good understanding of the theory and methodology of machine learning in quantitative investment and effective application of techniques to real-world financial data using Python. Shows a high level of ability to model and interpret financial patterns and risks with machine learning, exceeding expectations for high-quality research in empirical asset pricing and for industry roles.
B	Satisfactory to good understanding of the theory and methodology of machine learning in quantitative investment and solid application of techniques to financial data using Python. Demonstrates adequate ability to model and interpret financial patterns and risks using machine learning, meeting or slightly exceeding expectations for research in empirical asset pricing and for industry roles.
C	Basic understanding of the theory and methodology of machine learning in quantitative investment and limited application of techniques to financial data using Python. Shows basic ability to model and interpret financial patterns and risks with machine learning, meeting minimum requirements for research in empirical asset pricing and industry applications.
D	Limited understanding of the theory and methodology of machine learning in quantitative investment and minimal ability to apply techniques to financial data using Python. Lacks sufficient ability to model and interpret financial patterns and risks with machine learning to meet the standards for research or industry expectations.
F	Unsatisfactory performance on a number of learning outcomes, OR failure to meet specified assessment requirements.

8. Recommended but Optional Reading

Coqueret, G. and T. Guida (2020). *Machine Learning for Factor Investing*. Chapman and Hall. (available at https://www.mlfactor.com/)

Nagel, S. (2021). *Machine Learning in Asset Pricing*. Princeton University Press.

Zhang, M., T. Lu, and C. Shi (2025). *Navigating the Factor Zoo: The Science of Quantitative Investing*. Routledge.

石川，刘洋溢，连祥斌（2020）. 因子投资：方法与实践. 电子工业出版社

王熙，石川（2022）. 机器学习与资产定价. 电子工业出版社

Latest journal papers about asset pricing in general from *Journal of Finance*, *Journal of Financial Economics*, and *Review of Financial Studies*